Course Code: 23MAT204

Mathematics For Intelligent Systems III

Part- 8

Support Vector Machine

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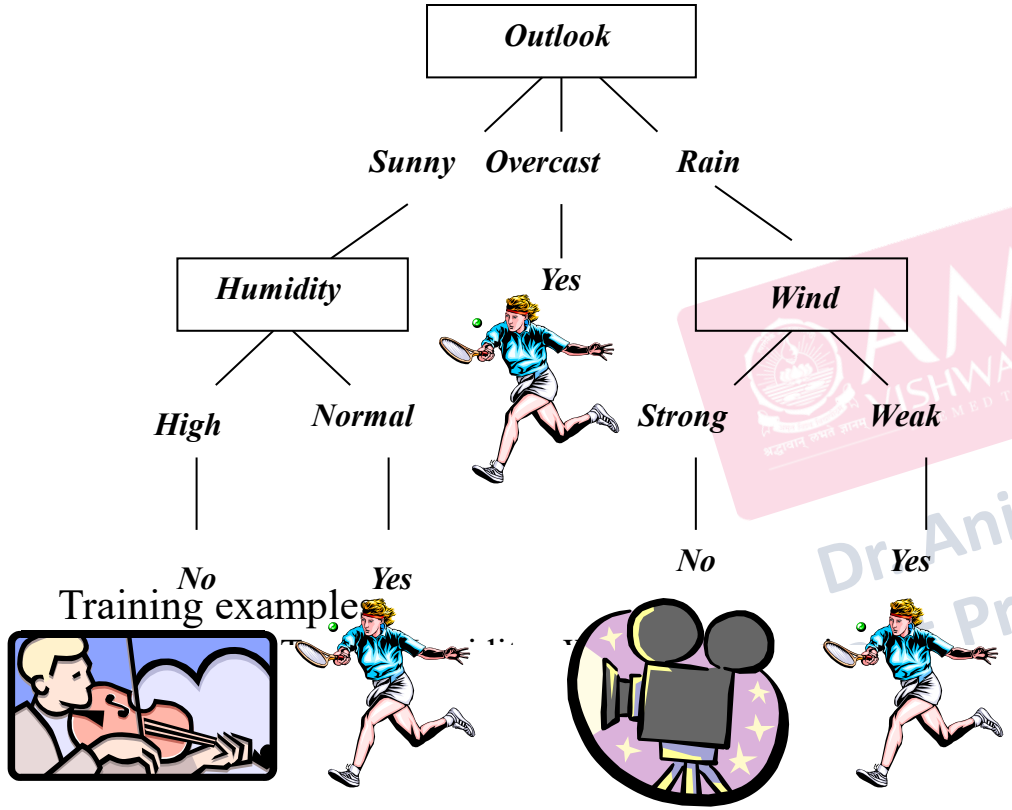
Classification

- Everyday, all the time we classify things.
- Eg crossing the street:
  - Is there a car coming?
  - At what speed?
  - How far is it to the other side?
  - Classification: Safe to walk or not!!!

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- **Decision tree learning**

IF (*Outlook = Sunny*) ^ (*Humidity = High*)
THEN *PlayTennis =NO*
IF (*Outlook = Sunny)^ (Humidity = Normal)*
THEN *PlayTennis = YES*



Training example

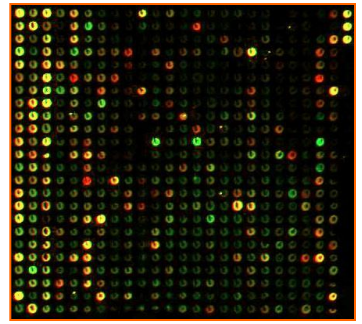
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Problems in classifying data

- Often high dimension of data.
- Hard to put up simple rules.
- Amount of data.
- Need automated ways to deal with the data.
- Use computers – data processing, statistical analysis, try to learn patterns from the data (Machine Learning)

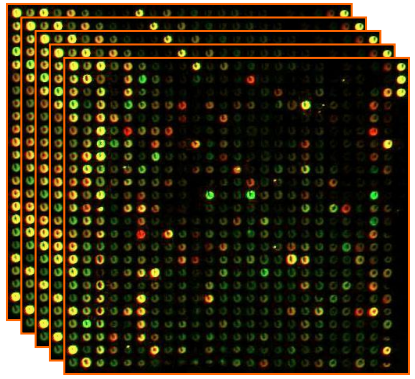
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Black box view of Machine Learning

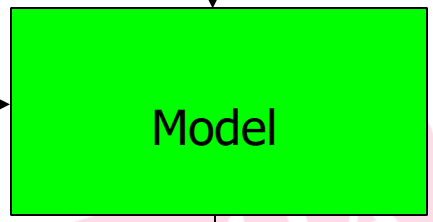


Test-data

Training data







Magic black box
(learning machine)

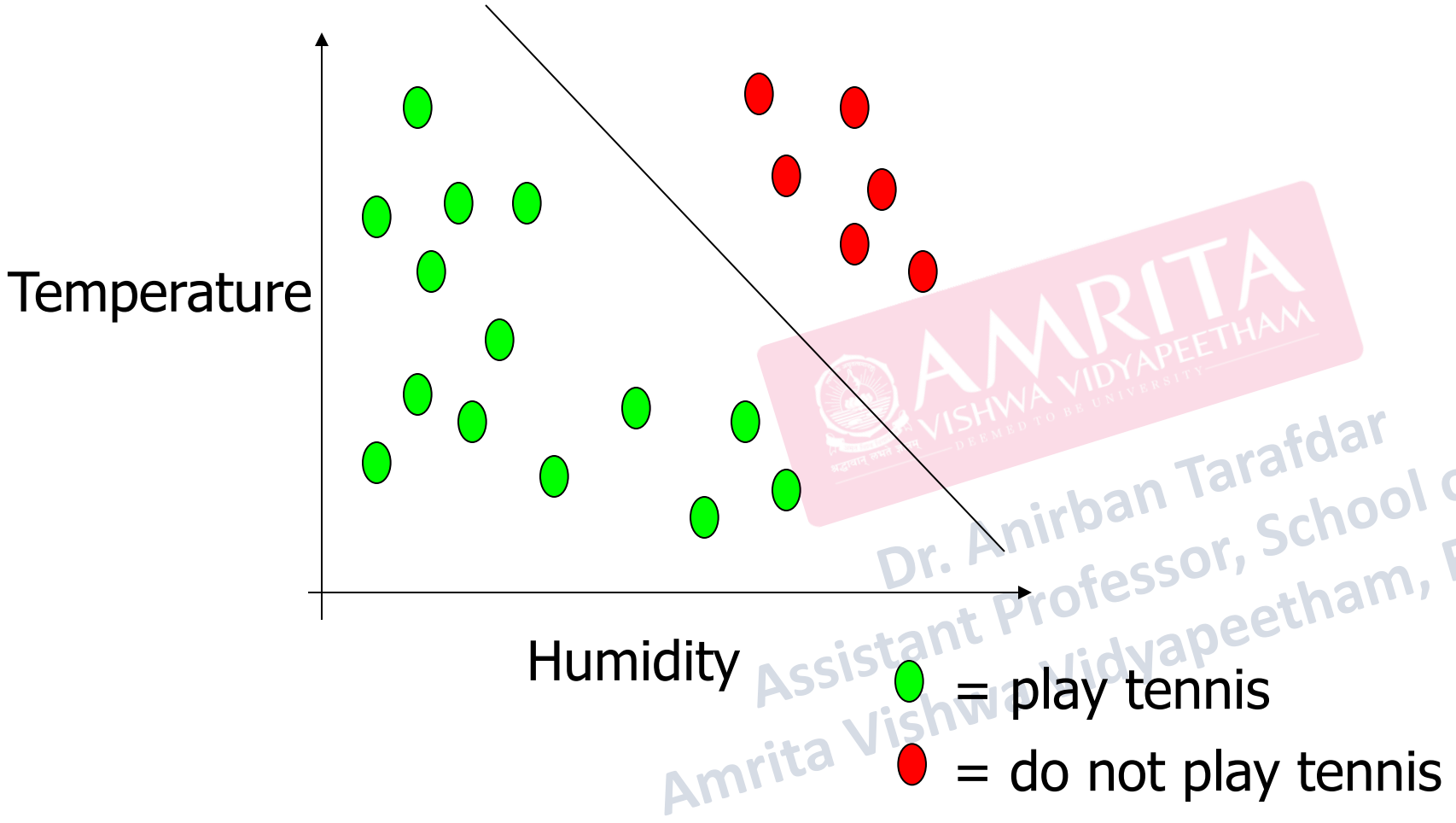
Prediction = cancer or not

Training data: -Expression patterns of some cancer + expression data from healty person

Model: - The model can distinguish between healty and sick persons. Can be used for prediction.

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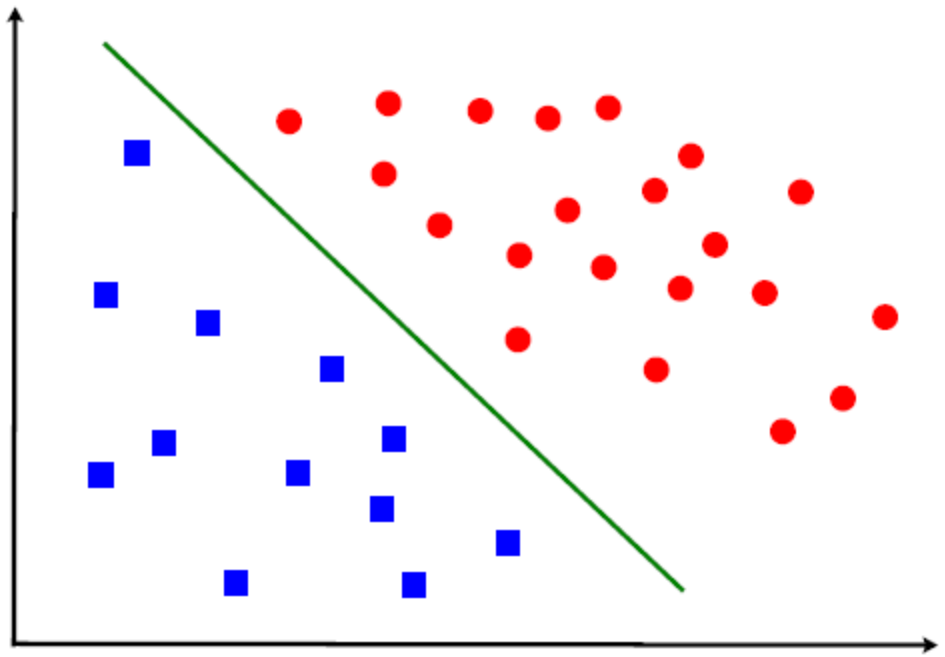
Tennis example 2



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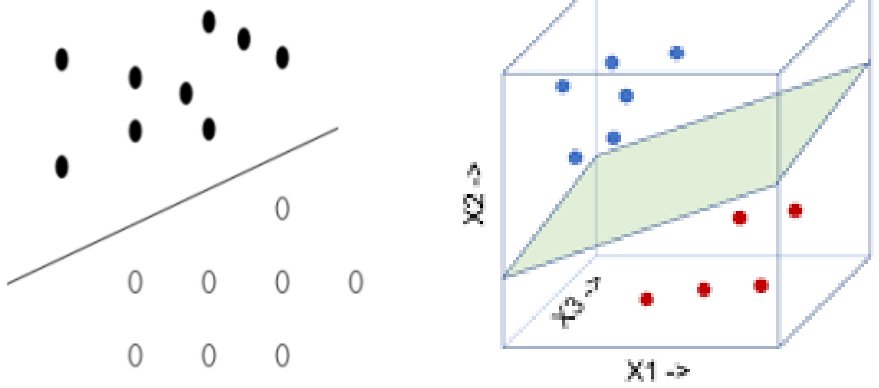
Linearly Separable Classes

$$Y = \text{sgn}[\mathbf{w}^\top \mathbf{x} + b > 0] = \begin{cases} +1 & \mathbf{w}^\top \mathbf{x} + b > 0 \\ -1 & \mathbf{w}^\top \mathbf{x} + b \leq 0 \end{cases}$$

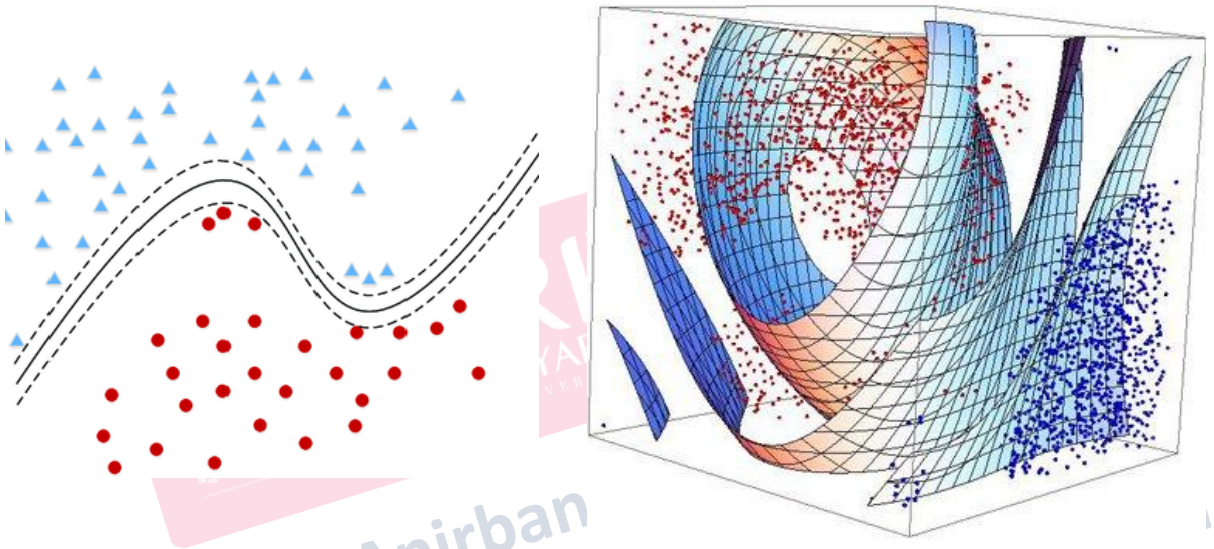


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Support vector machine – Linear or Non-linear



Linear SVM

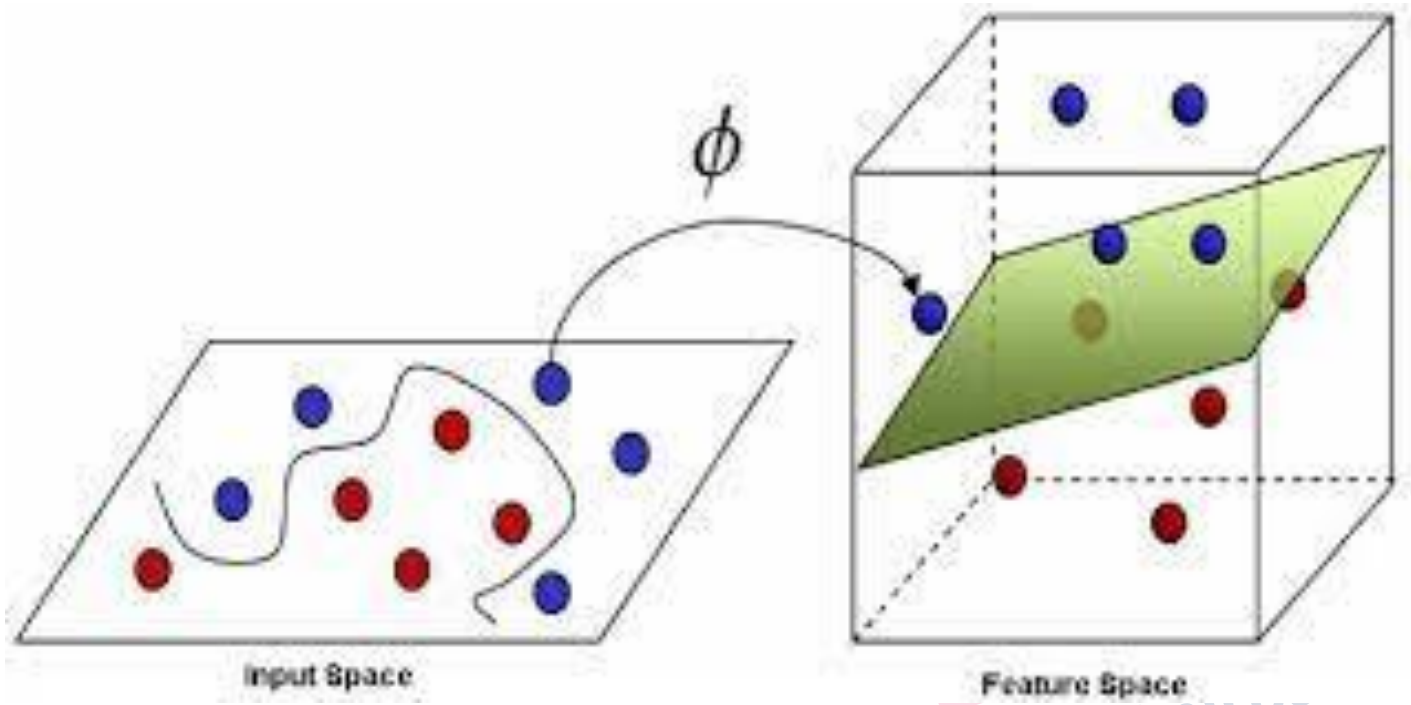


Non-Linear SVM

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[
Depending on the nature of the samples, the decision boundary can be linear or non-linear, resulting to linear SVM or non-linear SVM.
]

Non-Linear to Linear

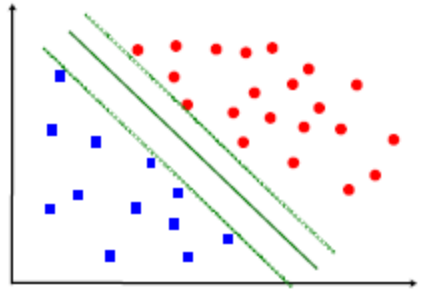


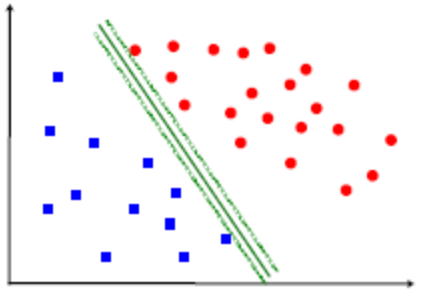
[
Finding a non-linear decision boundary is complex. Further, non-linear problems can be transformed to a linear problem. For example, ……. This lesson focusses on Linear SVM.
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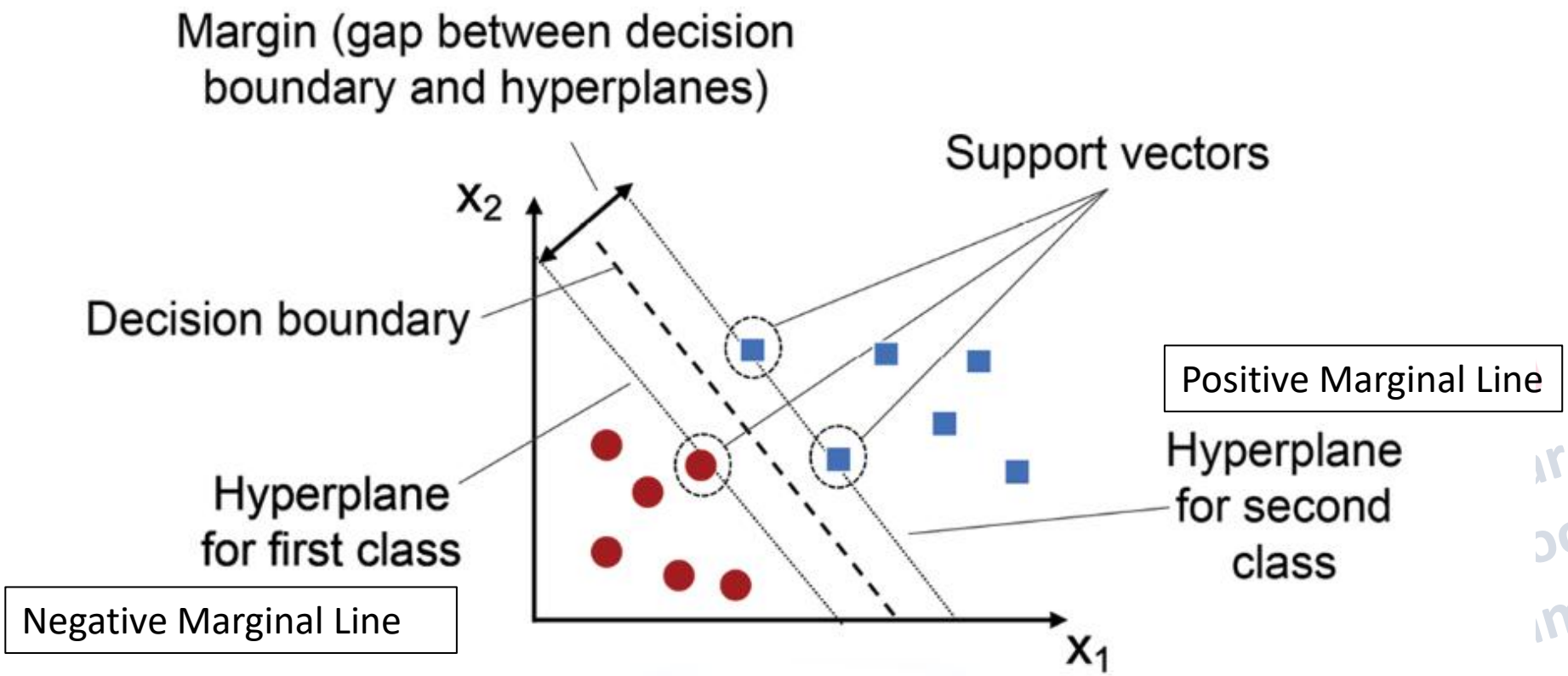
Selection of a Good Hyper-Plane

Objective: Select a `good' hyper-plane using only the data!
Intuition:
(Vapnik 1965) - assuming linear separability
(i) Separate the data
(ii) Place hyper-plane `far' from data



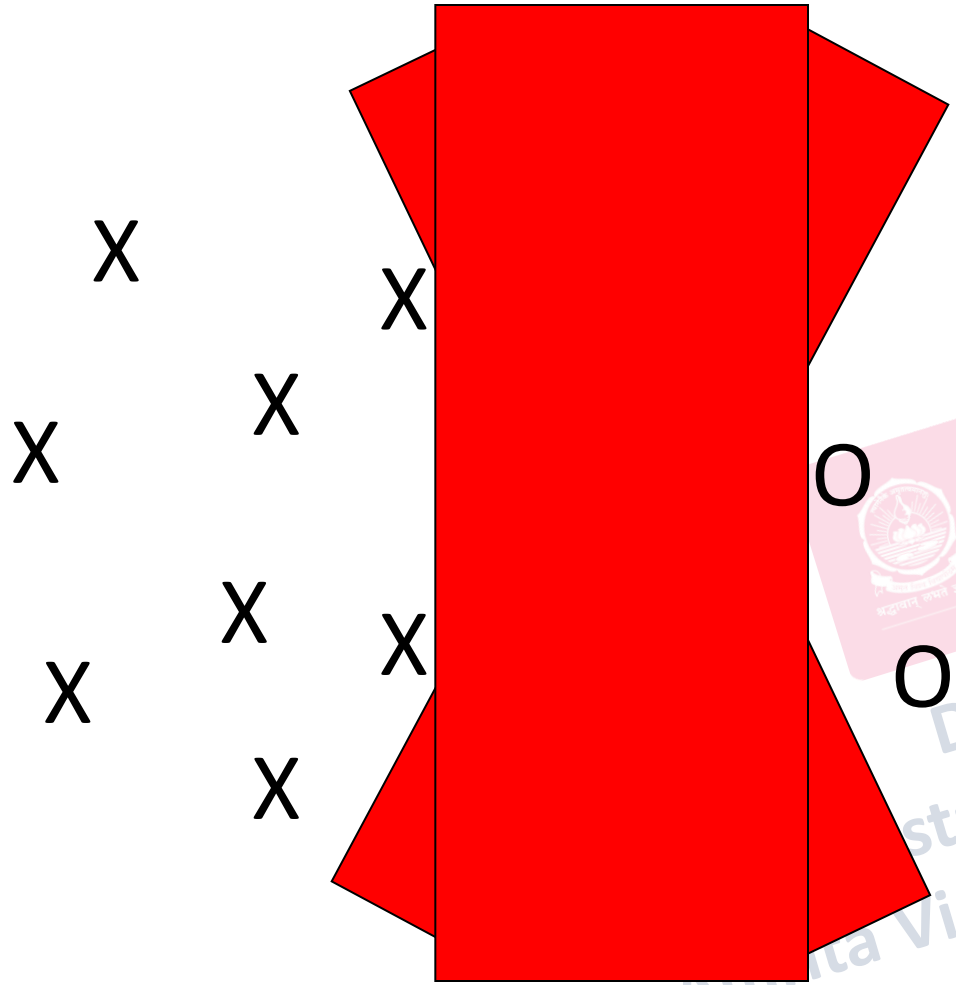


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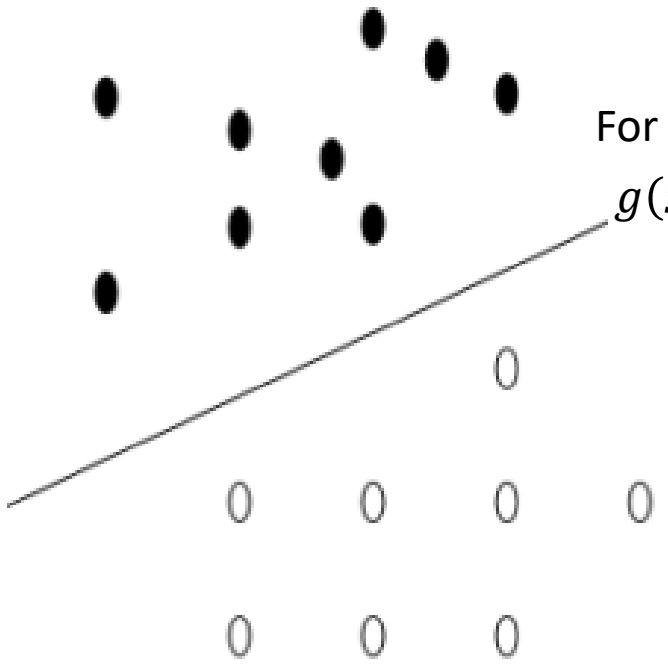
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“Fat” Separators



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Linear Support vector machine



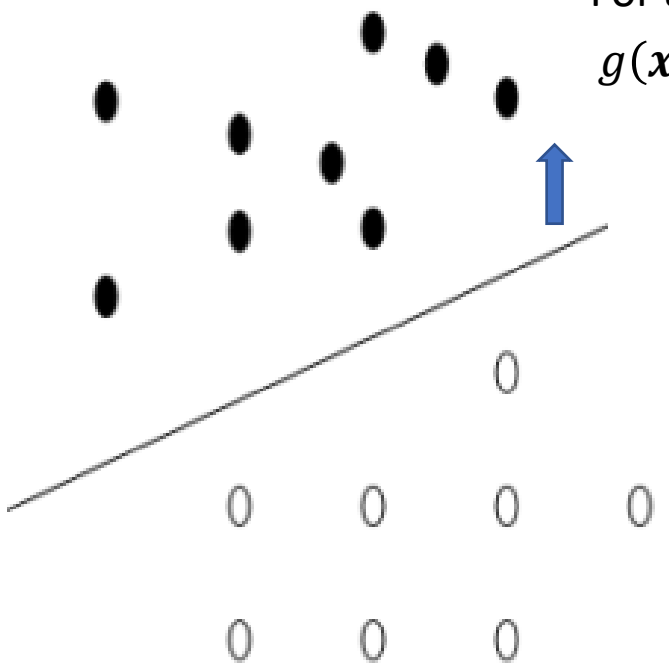
For all the points on the hyper plane, g(x) = 0

$$g(\boldsymbol{x}) = \boldsymbol{\omega}^T \boldsymbol{x} + \omega_0 = 0$$

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[
To understand the process, let us begin with as separating hyperplane defined by the linear function g(x).
]

Linear Support vector machine

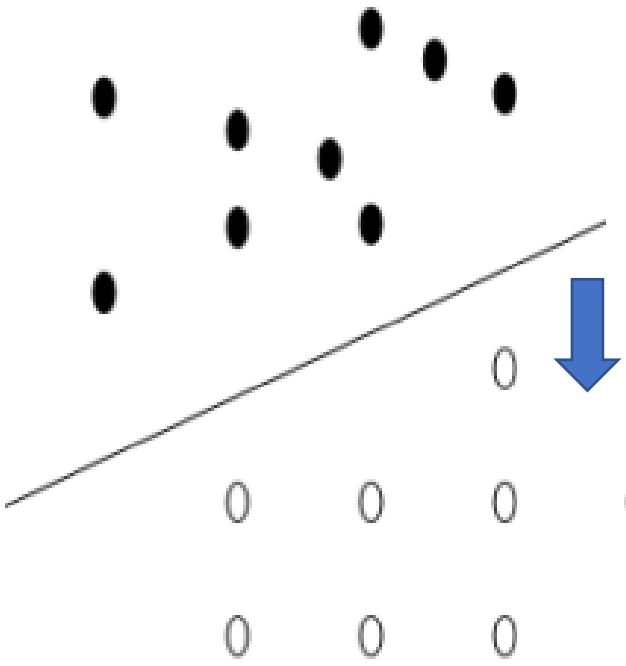


For all the points above the hyper plane, g(x) > 0

$$g(\boldsymbol{x}) = \boldsymbol{\omega}^T \boldsymbol{x} + \omega_0 > 0$$

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Linear Support vector machine

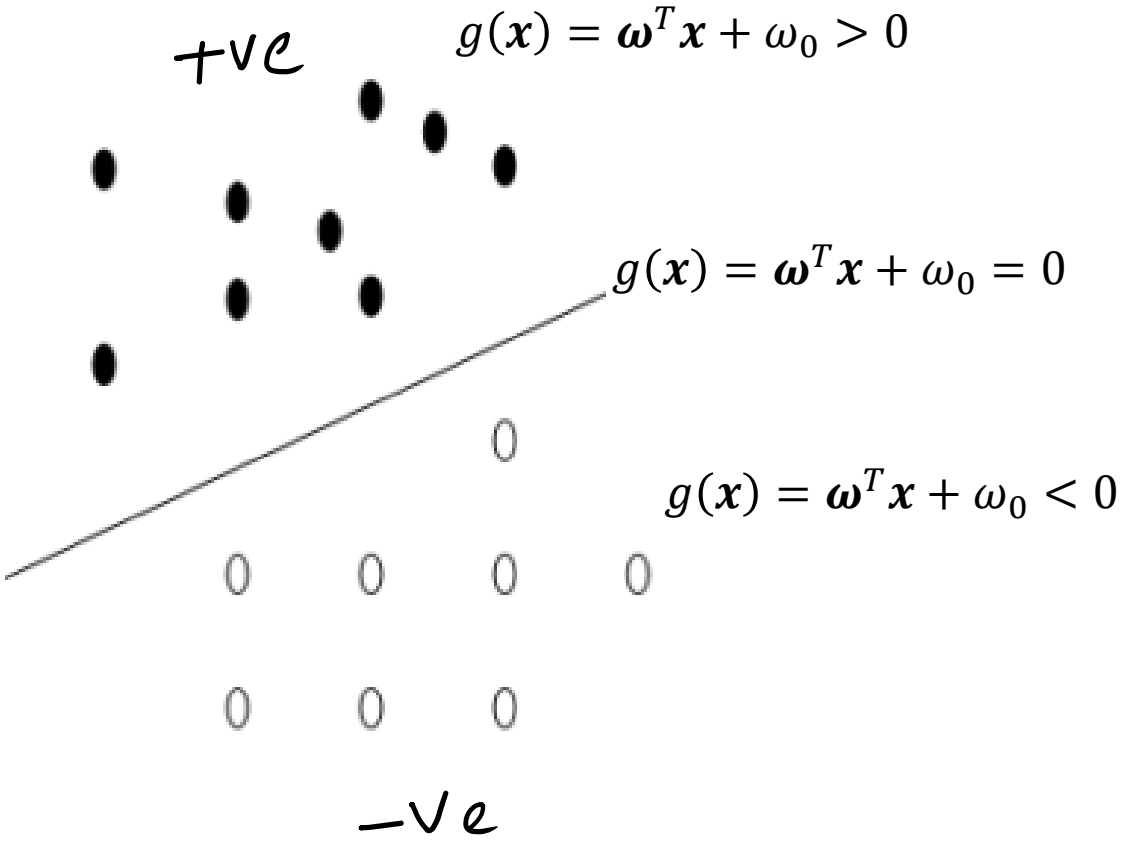


$$g(\boldsymbol{x}) = \boldsymbol{\omega}^T \boldsymbol{x} + \omega_0 < 0$$

For all the points below the hyper plane, g(x) < 0

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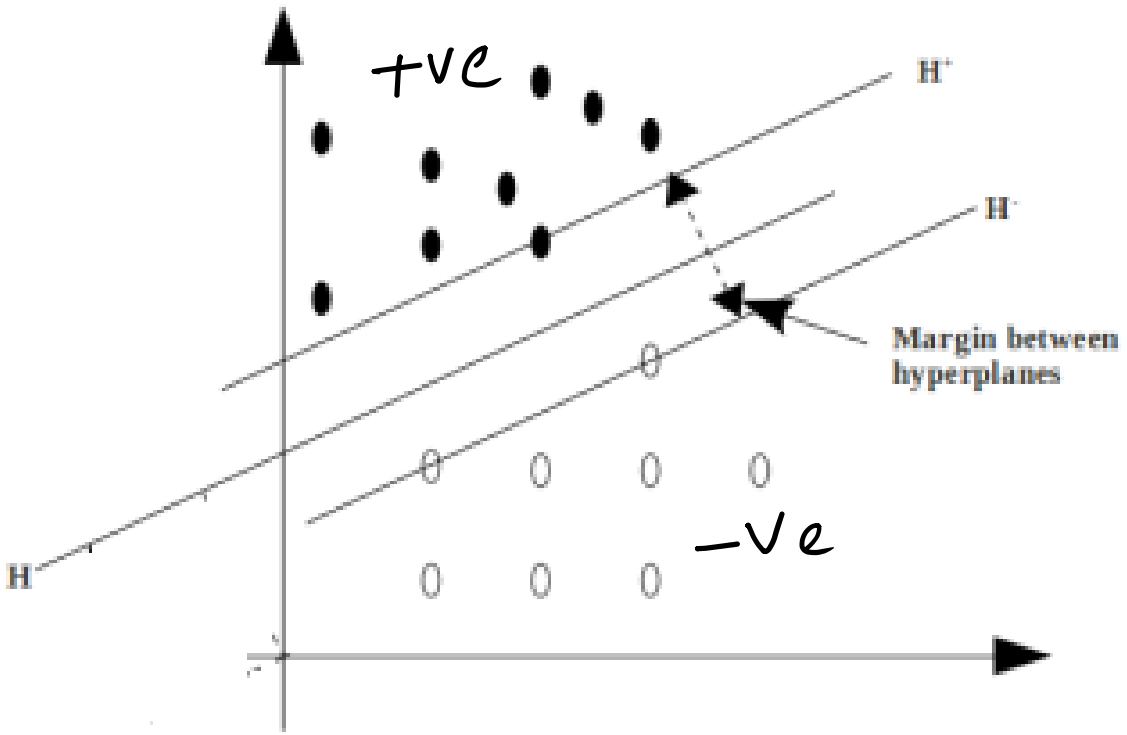
Linear Support vector machine



We choose the hyperplane that has the **maximum separating margin**

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What is Separating Margin?



H is the separating hyperplane

H^+ is the plane parallel to H and passing through the nearest +ve points to H

H^- is the plane parallel to H and passing through the nearest -ve points to H

Separating margin is the distance between H^+ and H^-

We choose H that has the maximum margin.

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Definitions

Define the hyperplane H such that:

$\mathbf{x}_i \bullet \mathbf{w} + b \geq +1$ when $y_i = +1$

$\mathbf{x}_i \bullet \mathbf{w} + b \leq -1$ when $y_i = -1$

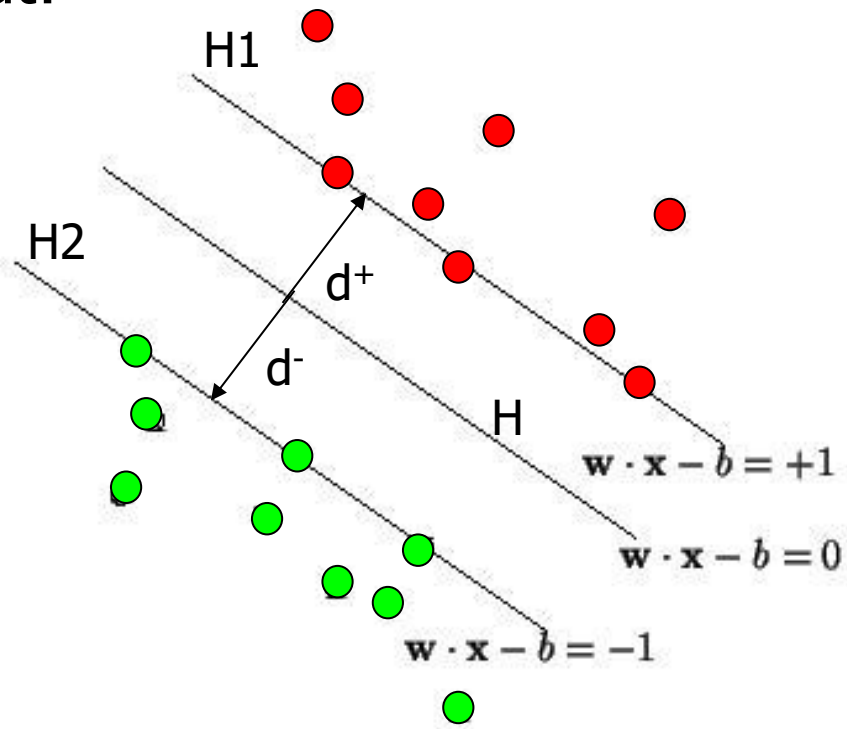
H1 and H2 are the planes:

H1: $\mathbf{x}_i \bullet \mathbf{w} + b = +1$

H2: $\mathbf{x}_i \bullet \mathbf{w} + b = -1$

The points on the planes H1 and H2 are the **Support Vectors**:

$$\{\mathbf{x}_i : |\mathbf{w}^\top \mathbf{x}_i + b| = 1\}$$



d+ = the shortest distance to the closest positive point

d- = the shortest distance to the closest negative point

The margin of a separating hyperplane is $d^+ + d^-$.

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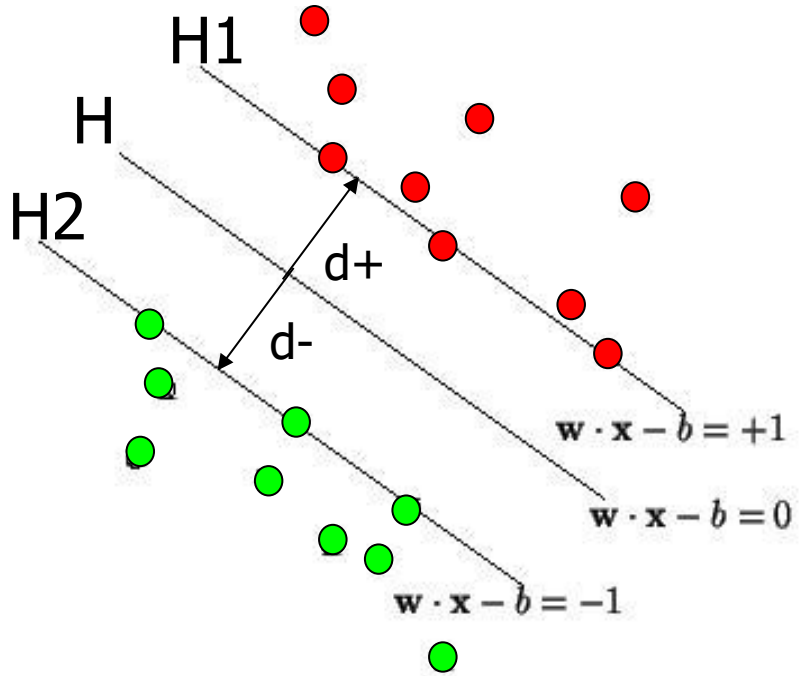
Maximizing the margin

We want a classifier with as big margin as possible.

Recall the distance from a point(x_0,y_0) to a line:
Ax+By+c = 0 is|A x_0 +B y_0 +c|/sqrt(A^2+B^2)

The distance between H and H1 is:
|**w•x**+b|/||w||=1/||w||

The distance between H1 and H2 is: 2/||w||



In order to maximize the margin, we need to minimize ||w||. With the condition that there are no datapoints between H1 and H2:

$\mathbf{x}_i\bullet\mathbf{w}+b \geq +1$ when $y_i = +1$
$\mathbf{x}_i\bullet\mathbf{w}+b \leq -1$ when $y_i = -1$

Can be combined into $y_i(\mathbf{x}_i\bullet\mathbf{w}) \geq 1$

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Optimization Problem

Distance from support vector to $H(\mathbf{w}, b)$

$$\frac{\mathbf{w}^\top \mathbf{x}_i + b}{\|\mathbf{w}\|} = \frac{\pm 1}{\|\mathbf{w}\|}$$

$$\textbf{Margin} = \left| \frac{1}{\|\mathbf{w}\|} - \frac{-1}{\|\mathbf{w}\|} \right| = \frac{2}{\|\mathbf{w}\|}$$

minimize	$\frac{1}{2}\mathbf{w}^\top \mathbf{w}$
subject to	$y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 \qquad i = 1, 2, \ldots, n$

1. Convex quadratic program
2. Linear inequality constraints (many!)
3. $d + 1$ parameters, n constraints

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The Lagrangian trick

Reformulate the optimization problem: A ”trick” often used in optimization is to do an Lagrangian formulation of the problem. The constraints will be replaced by constraints on the Lagrangian multipliers and the training data will only occur as dot products.

Objective function is to **minimize** $\frac{\boldsymbol{\omega}^T\boldsymbol{\omega}}{2}$

Subject to $y_i(\boldsymbol{\omega}^T\boldsymbol{x_i} + \omega_0) \geq 1, \forall \boldsymbol{x_i}$

To find the parameters $\boldsymbol{\omega}$ and ω_0, solve the following optimization function

$$\boldsymbol{L_p} = \frac{\boldsymbol{\omega}^T\boldsymbol{\omega}}{2} - \sum_{i=1}^{n} \lambda_i(y_i(\boldsymbol{\omega}^T\boldsymbol{x_i} + \omega_0) - 1)$$

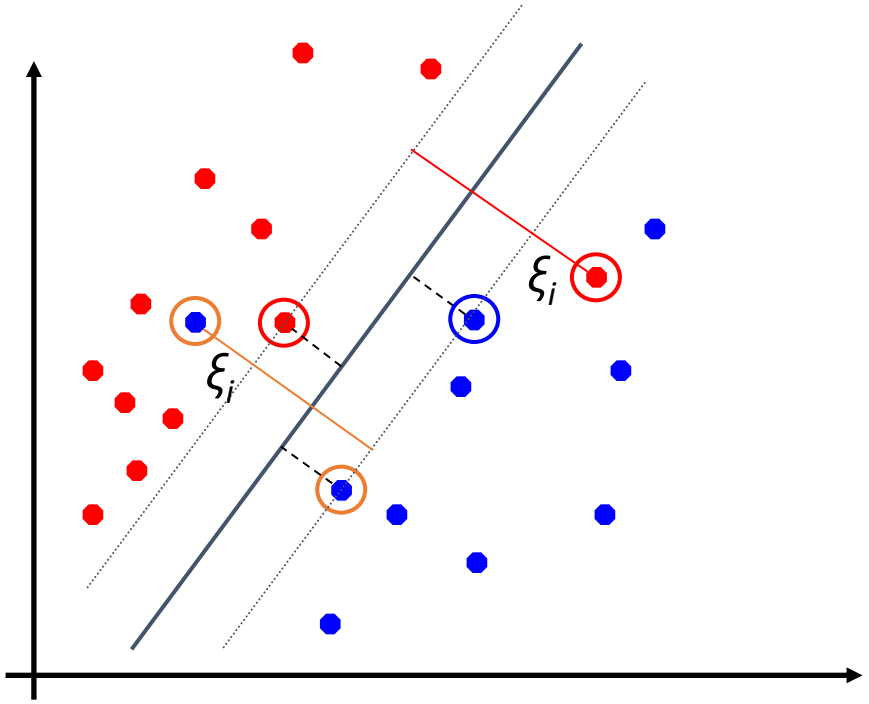
Where λ_i are Lagrange multipliers

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Soft Margin Classification

- What if the training set is not linearly separable?
- *Slack variables* ξ_i can be added to allow misclassification of difficult or noisy examples, resulting margin called *soft*.



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Non-Separable Case

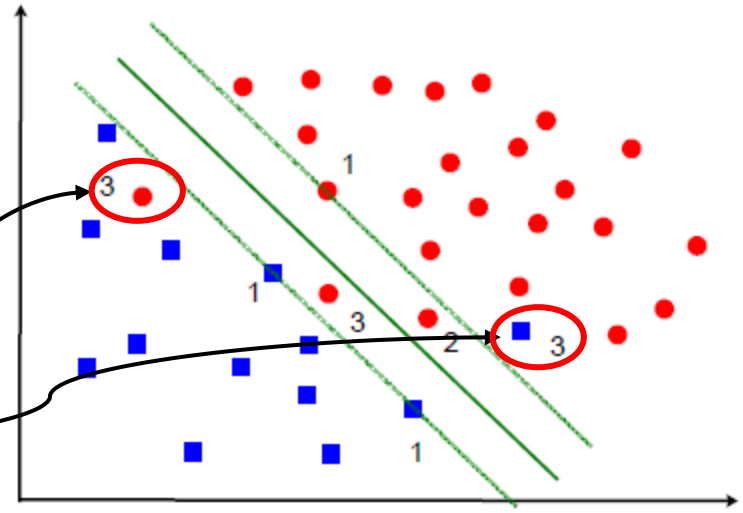
Objective: find a good separating hyper-plane for the non-separable case

Problem: Cannot satisfy $y_i[\mathbf{w}^\top \mathbf{x}_i + b] \geq 1$ for all i

Solution: *Slack* variables

$$\mathbf{w}^\top \mathbf{x}_i + b \geq +1 - \xi_i \quad \text{for } y_i = +1,$$
$$\mathbf{w}^\top \mathbf{x}_i + b \leq -1 + \xi_i \quad \text{for } y_i = -1,$$
$$\xi_i \geq 0 \quad k = 1, 2, \ldots, n.$$

An error occurs if $\xi_i > 1$. Thus,



$$\sum_{i=1}^{n} I(\xi_i > 1) = \#\text{ errors}$$

We need to minimize these points too

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Proposed solution: Minimize

$$\frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n} I(\xi_i > 1) \qquad (\text{non} - \text{convex!})$$

Suggestion: Replace $I(\xi_i > 1)$ by ξ_i (upper bound)

$$\begin{aligned} \underset{w,b,\xi}{\text{minimize}} \quad & L_P(\mathbf{w},\xi) = \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{n}\xi_i \\ \text{subject to} \quad & y_i(\mathbf{w}^\top \mathbf{x}_i + b) \geq 1 - \xi_i \\ & \xi_i \geq 0 \end{aligned}$$

Tradeoff: Large C - penalize errors, Small C penalize complexity

Dual Problem: Same as in separable case, except that $0 \leq \alpha_i \leq C$

Support vectors: $\alpha_i > 0$ - but lose geometric interpretation!

Solution:

$$\mathbf{w} = \sum_{i=1}^{n} \alpha_i y_i \mathbf{x}_i$$

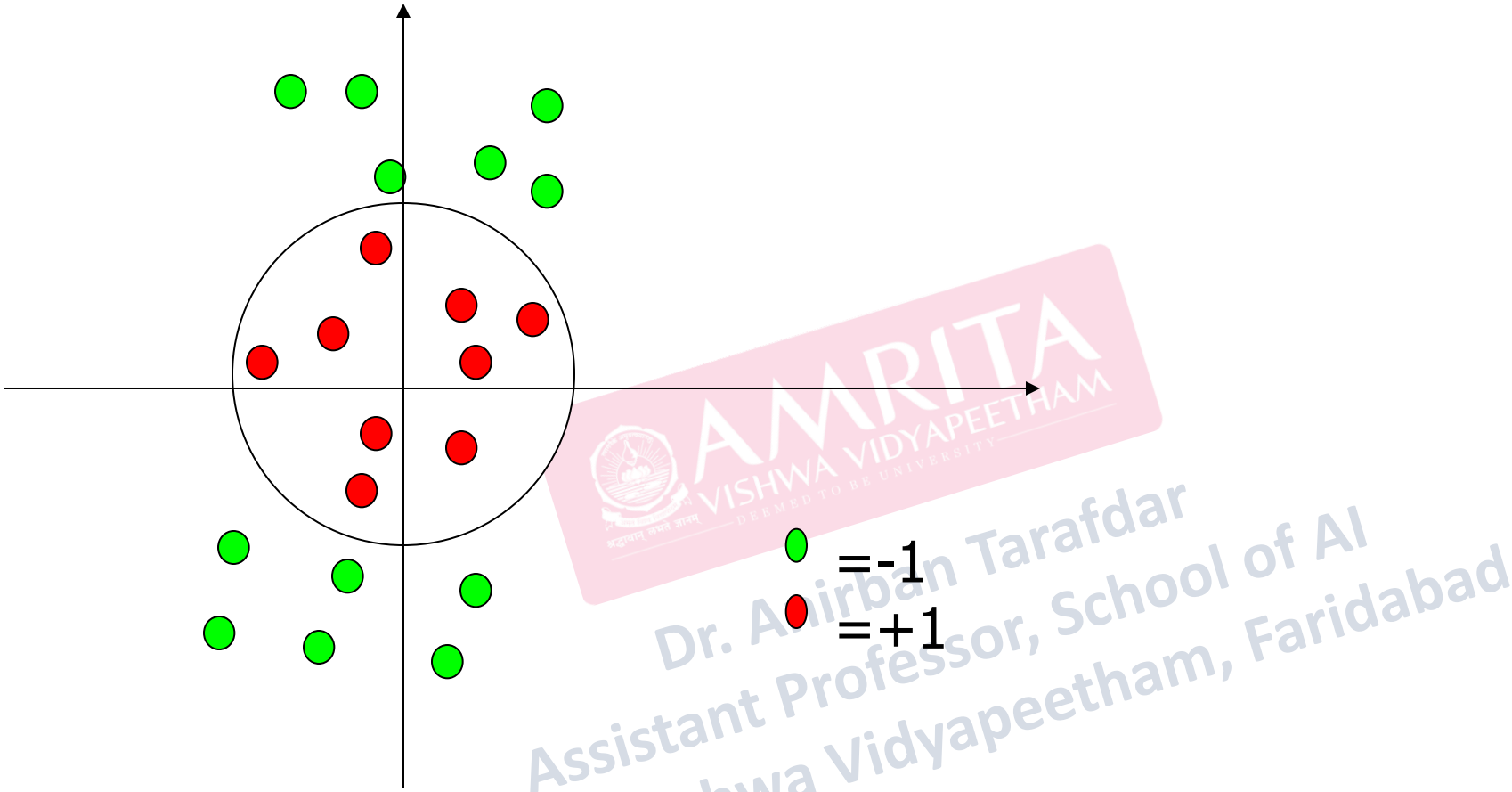
$$f(\mathbf{x}) = \text{sgn}\left(\sum_i \alpha_i y_i \mathbf{x}_i^\top \mathbf{x} + b\right)$$

KKT conditions:

$$\begin{aligned} 0 &= \sum_{i=1}^{n} \alpha_i y_i \\ 0 &= \alpha_i(y_i\left(\mathbf{w}^\top \mathbf{x}_i + b\right) - 1 + \xi_i) \\ 0 &= (C - \alpha_i)\xi_i \end{aligned}$$

Support vectors: characterized by $\alpha_i > 0$

Problems with linear SVM

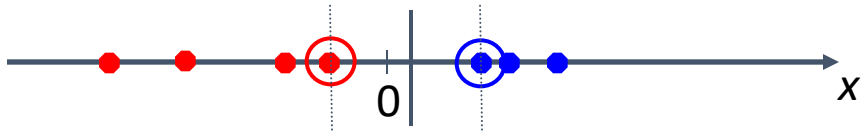


What if the decison function is not a linear?

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Non-linear SVMs

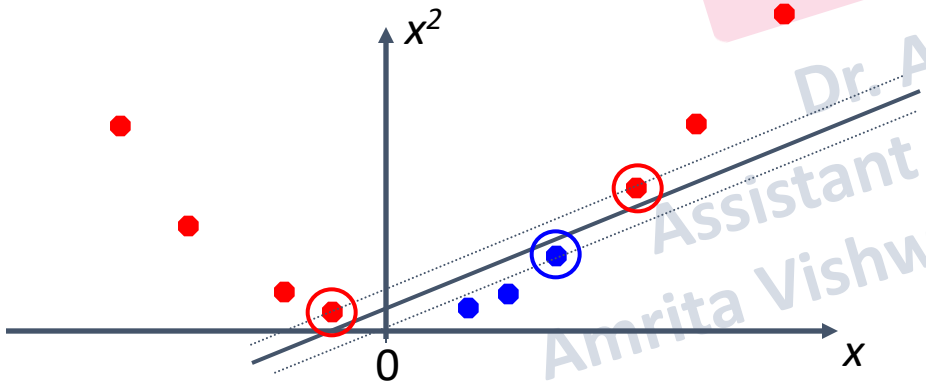
Datasets that are linearly separable with some noise work out great:



- But what are we going to do if the dataset is just too hard?



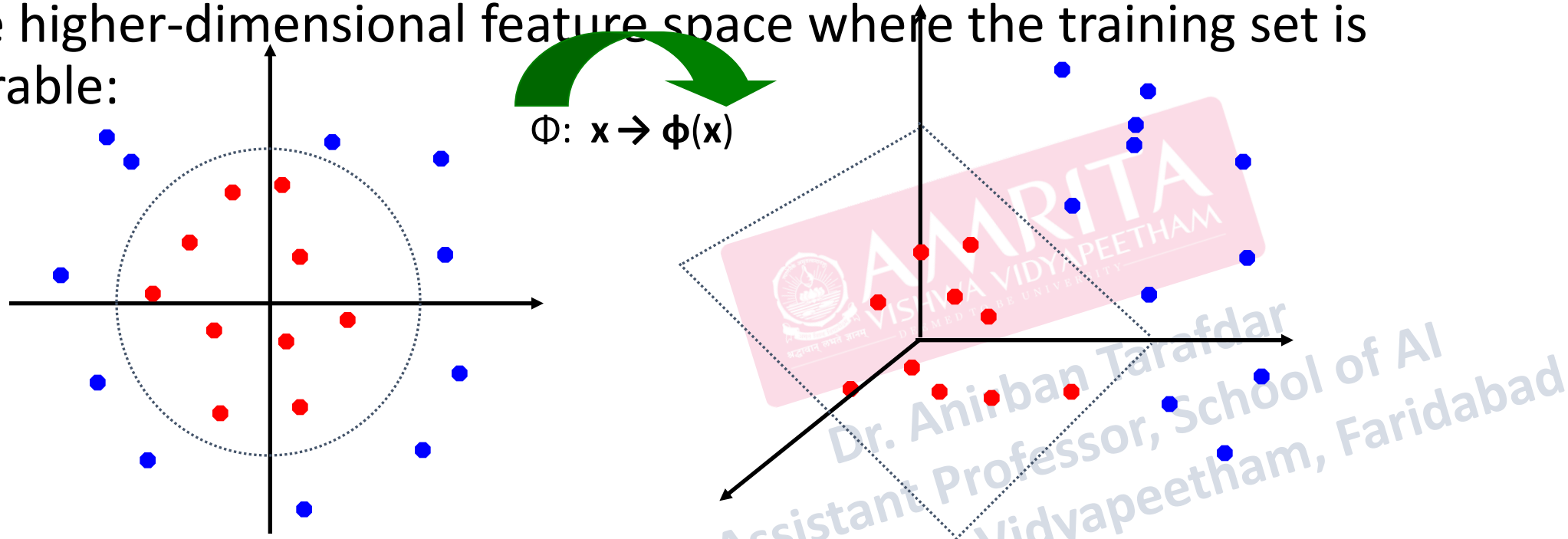
- How about… mapping data to a higher-dimensional space:



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Non-linear SVMs: Feature spaces

- General idea: the original feature space can always be mapped to some higher-dimensional feature space where the training set is separable:



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Examples of Kernel Functions

- Linear: $K(\mathbf{x}_i,\mathbf{x}_j)= \mathbf{x}_i^T\mathbf{x}_j$
  - Mapping Φ: $\mathbf{x} \rightarrow \boldsymbol{\phi}(\mathbf{x})$, where $\boldsymbol{\phi}(\mathbf{x})$ is $\mathbf{x}$ itself

- Polynomial of power p: $K(\mathbf{x}_i,\mathbf{x}_j)= (1+ \mathbf{x}_i^T\mathbf{x}_j)^p$

- Gaussian (radial-basis function): $K(\mathbf{x}_i,\mathbf{x}_j) = e^{-\frac{\|\mathbf{x}_i-\mathbf{x}_j\|^2}{2\sigma^2}}$
  - Mapping Φ: $\mathbf{x} \rightarrow \boldsymbol{\phi}(\mathbf{x})$, where $\boldsymbol{\phi}(\mathbf{x})$ is *infinite-dimensional*: every point is mapped to *a function* (a Gaussian); combination of functions for support vectors is the separator.

- Higher-dimensional space still has *intrinsic* dimensionality *d* (the mapping is not *onto*), but linear separators in it correspond to *non-linear* separators in original space.

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SVM applications

- SVMs were originally proposed by Boser, Guyon and Vapnik in 1992 and gained increasing popularity in late 1990s.
- SVMs are currently among the best performers for a number of classification tasks ranging from text to genomic data.
- SVMs can be applied to complex data types beyond feature vectors (e.g. graphs, sequences, relational data) by designing kernel functions for such data.
- SVM techniques have been extended to a number of tasks such as regression [Vapnik *et al.* '97], principal component analysis [Schölkopf *et al.* '99], etc.
- Most popular optimization algorithms for SVMs use *decomposition* to hill-climb over a subset of α_i's at a time, e.g. SMO [Platt '99] and [Joachims '99]
- Tuning SVMs remains a black art: selecting a specific kernel and parameters is usually done in a try-and-see manner.

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